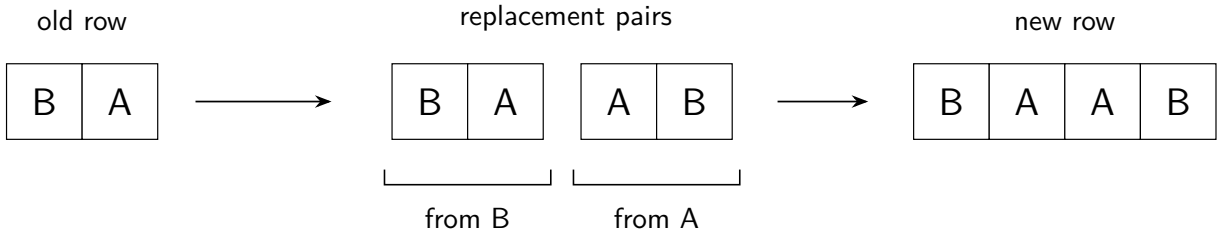
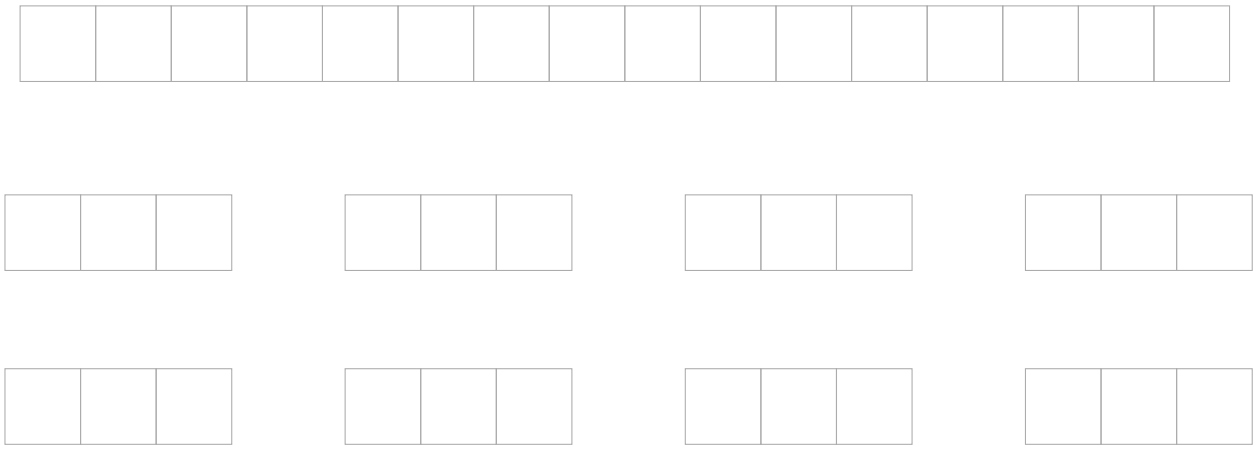
Replace A by AB and B by BA. Replace each old tile once. Keep the old and new rows separate while your partner checks.

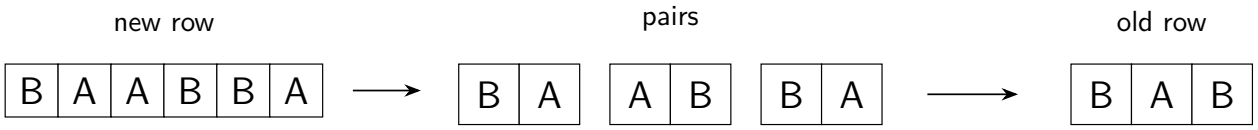


Problem 1: Grow a row of 16 tiles from A. Find every different strip of three neighboring tiles in your row. Record each short strip once.



Problem 2: Could AAA or BBB ever appear in a longer row grown from A? Explain why, or build a row that contains one.

Problem 3: Undo a replacement by turning each pair AB into A and each pair BA into B.



Each row below claims to be a whole row grown from a single A. Which claims can be right? Show what settles each claim.

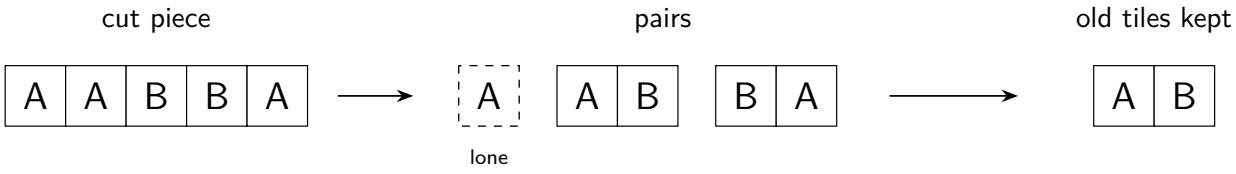
A	B	B	A	B	A	A	B

A	B	B	A	A	B	B	A

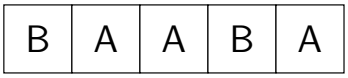
A	B	B	A	B	B	A	B

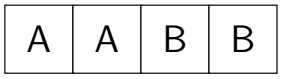
A	B	B	A	B	A	A	B	A	B	B	A	B	A	A	B

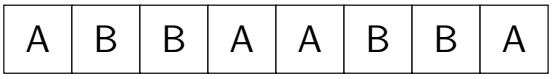
Problem 4: Each piece below was cut from a longer row grown from A. Find every way to pair the tiles as AB or BA. Use every tile, apart from at most one lone tile at each end. Decode only complete pairs.











Problem 5: Cut a piece with five tiles from a row grown from A. Can two different ways of pairing it, as in Problem 4, fit the replacement rule? Find such a piece, or explain why none exists.

Problem 6: One strip repeats AB forever. Another repeats ABBA forever. For each strip, find a finite piece that cannot occur in any row grown from A. Explain why your piece is impossible.

A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	⋯

A	B	B	A	A	B	B	A	A	B	B	A	A	B	B	A	⋯

Problem 7: Imagine the endless strip made by growing from A. Could everything after some point repeat one fixed block forever? Explain why, or describe a block and where the repeating part starts.